

Web Browsing, Search, Upload & Download

Lesson 36 — Search, Storage and Transfer

JKSSB Computer Awareness

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What a Web Browser Does

- ▶ A **web browser** is the software used to open and view web pages, such as Google Chrome, Mozilla Firefox, and Microsoft Edge.
- ▶ The browser sends a request for a page and displays the returned text, images, and links.
- ▶ The page address typed in the **address bar** is called the **URL** — Uniform Resource Locator.
- ▶ Each open page lives in its own **tab**, so several pages can stay open at once.

The one idea

The browser fetches pages; the search engine finds pages; tabs keep several pages open together.

Tabs, Bookmarks, and History

- ▶ **Tabs** let many pages stay open in one window; a new tab opens with **Ctrl+T**.
- ▶ A **bookmark** is a saved shortcut to a page so it can be opened again without typing the address.
- ▶ **Browsing history** is the list of pages visited earlier, stored by date.
- ▶ History answers “where was I”; a bookmark answers “where do I want to return”.

Office desktop example

While preparing a report, the clerk keeps the office portal in one tab and the search results in another, and bookmarks the portal for tomorrow.

How a Search Engine Works: Crawling

- ▶ A **search engine** (for example Google, Bing) finds pages that match the typed words.
- ▶ Step 1 — **crawling**: automatic programs called **crawlers** or spiders visit pages and follow their links.
- ▶ Crawling discovers new and changed pages across the web.
- ▶ Without crawling, the engine would never learn that a new page exists.

Pipeline position

Crawling is the collection step: it gathers pages before anything else can happen.

Indexing and Ranking

- ▶ Step 2 — **indexing**: the crawled pages are stored in a large **index**, like the index at the back of a book.
- ▶ Step 3 — **ranking**: when words are typed, the engine picks matching pages from the index and orders them by relevance.
- ▶ The ordered list of matches is the **search results page**.
- ▶ Order to remember: crawl, then index, then rank.

Trap alert

The index is built **before** the search; ranking happens only after the words are typed.

Sponsored Results vs Organic Results

- ▶ **Organic results** are the normal ranked matches chosen by relevance.
- ▶ **Sponsored results** (labelled Ad / Sponsored) are paid placements shown because an advertiser paid, not because they ranked highest.
- ▶ A sponsored result always carries a visible **Ad** or **Sponsored** label.
- ▶ Being listed first on the page does **not** always mean being the most relevant match.

Keep the pair straight

Organic = ranked by relevance; sponsored = placed by payment.

Cookies: What They Remember

- ▶ A **cookie** is a small file a website stores on the computer to remember that visitor.
- ▶ Cookies keep a user logged in, remember items in a cart, and store preferences such as language.
- ▶ They are set **by websites** and read back on the next visit.
- ▶ Deleting cookies signs the user out of sites and clears saved preferences.

One-line definition

A cookie remembers **who the visitor is** for the website.

Cache: What It Stores

- ▶ The browser **cache** stores **copies of page parts** — images, styles, and scripts — on the computer.
- ▶ On the next visit the saved copies load from the disk instead of being fetched again, so the page opens **faster**.
- ▶ Cache is managed **by the browser**, not by the website.
- ▶ Clearing the cache does not sign the user out; it only makes the next visit slightly slower.

One-line definition

The cache remembers **page parts** so repeat visits load faster.

Cookies vs Cache vs History

Item	What it stores	Who stores it
Cookie	Visitor data (login, cart, choices)	Website
Cache	Copies of page files (images, styles)	Browser
History	List of visited page addresses	Browser

Trap alert

Cookie = who you are; cache = saved page parts; history = where you went. Do not swap the three.

Downloading

- ▶ **Downloading** means copying a file **from the internet to the computer**.
- ▶ Examples: saving a form PDF from the office portal, saving a photo from a web page.
- ▶ A download moves data **down** to the user's machine and is measured by **download speed**.
- ▶ The saved file then opens without needing the internet.

Office desktop example

The clerk downloads the leave-application PDF from the portal to the desktop, then fills it offline in a reader.

Uploading

- ▶ **Uploading** means sending a file **from the computer to the internet**.
- ▶ Examples: attaching a document to an email, submitting a scanned form on a portal.
- ▶ An upload moves data **up** from the user's machine to a server.
- ▶ Sending the file needs the internet connection at sending time.

Trap alert

Download = internet to computer; upload = computer to internet. The direction is always judged from the user's machine.

Download Speed vs Upload Speed

Point	Download	Upload
Direction	Server to computer	Computer to server
Use	Opening pages, saving files	Sending mail attachments, submitting forms
Typical speed	Faster	Slower

Home and office plans usually give a faster download than upload.

File Conversion

- ▶ **File conversion** means changing a file from one format to another, keeping the same content.
- ▶ Common office need: converting a Word document (**.docx**) to **PDF** before submitting it on a portal.
- ▶ Conversion changes the **format**, not the transfer direction: a converted file must still be uploaded or downloaded separately.
- ▶ The file **extension** (such as .pdf, .docx, .jpg) names the format the file is stored in.

Extension vs format

The extension is the name tag (.png); the format is the storage method (PNG). Do not conflate them.

Safe and Correct Practice

- ▶ Download files only from trusted sites; a free download can carry a **virus**.
- ▶ Check the file extension before opening: a document should not arrive as an unknown program file.
- ▶ Use **Ctrl+D** to bookmark the current page and **Ctrl+H** to open browsing history.
- ▶ Clear history to hide visited addresses; clear cookies to sign out of sites.

Trap Alert: Facts to Keep Straight

- ▶ Search order: **crawl**, then **index**, then **rank**.
- ▶ Sponsored = paid placement with an **Ad** label; organic = ranked by relevance.
- ▶ Cookie (website, visitor data) vs cache (browser, page copies) vs history (browser, visited list).
- ▶ Download = to the computer; upload = from the computer to the server.
- ▶ Conversion changes the **format**; transfer changes the **location**.

Lesson 36: Recall

- ▶ Browsers fetch pages; tabs hold several pages; bookmarks save shortcuts; history lists visited pages.
- ▶ Search pipeline: crawlers collect pages, the index stores them, ranking orders the matches; sponsored results are paid Ads.
- ▶ Cookies remember the visitor (website); cache keeps page copies (browser); history records visited addresses (browser).
- ▶ Downloading copies from the internet to the computer; uploading sends from the computer to the internet.
- ▶ File conversion changes the format (for example .docx to PDF); the extension names that format.